Mini project report on

Personal Credit Health and Debt Repayment Planning System

Submitted in partial fulfilment of the requirements for the award of degree of

Bachelor of Technology in Computer Science & Engineering UE24CS351A – DBMS Project

Submitted by:

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under the guidance of

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FACULTY OF ENGINEERING

PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)

Electronic City, Hosur Road, Bengaluru – 560 100, Karnataka, India

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CERTIFICATE

This is to certify that the mini project entitled

Personal Credit Health and Debt Repayment Planning System

is a bonafide work carried out by

Amoghaa R	**PES2UG24CS625**
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In partial fulfilment for the completion of fifth semester DBMS Project (UE24CS351A) in the Program of Study Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period AUG. 2026 – DEC. 2026. It is certified that all corrections / suggestions indicated for internal assessment have been incorporated in the report. The project has been approved as it satisfies the 5th semester academic requirements in respect of project work.

Signature
Dr. Shilpa S
Assistant Professor

DECLARATION

We hereby declare that the DBMS Project entitled **Personal Credit Health and Debt Repayment Planning System** has been carried out by us under the guidance of **Dr. Shilpa S, Assistant Professor** and submitted in partial fulfilment of the course requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester AUG – DEC 2026.

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ABSTRACT

The Personal Credit Health and Debt Repayment Planning System is a database-oriented project designed to centralize personal financial information and support effective debt repayment management.

The system maintains information related to users, financial profiles, loans, repayments, credit scores and repayment plans. A relational database is designed using an Entity Relationship model and implemented using MySQL.

The project demonstrates database concepts through DDL and DML statements, primary and foreign key constraints, SQL queries, transactions, stored procedures, triggers and indexing.

The proposed system provides a structured approach to managing financial records, monitoring repayment activities and organizing repayment plans. It demonstrates how database management concepts can be applied to a real-world personal financial management problem.

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1. INTRODUCTION

Personal financial management involves maintaining information about income, expenses, bank accounts, loans, credit scores and repayment activities. When this information is maintained separately, individuals may find it difficult to understand their overall financial position and manage repayment obligations effectively.

The Personal Credit Health and Debt Repayment Planning System is designed to provide a centralized database for organizing and managing personal financial information. It maintains user details, financial profiles, loan records, repayment history and repayment plans.

The system uses a relational database implemented in MySQL. Primary keys, foreign keys and integrity constraints maintain consistency between related records. SQL queries and database operations support financial analysis, repayment tracking and the management of repayment plans.

Objectives:

1. To maintain centralized personal financial records.
2. To store and manage loan and repayment information.
3. To monitor repayment status and outstanding amounts.
4. To maintain financial and credit-related information.
5. To support the creation and management of repayment plans.
6. To demonstrate database operations and advanced SQL concepts.

2. PROBLEM DEFINITION

Individuals often face difficulties managing their financial information because income, expenses, loans, credit scores and repayment records are maintained separately.

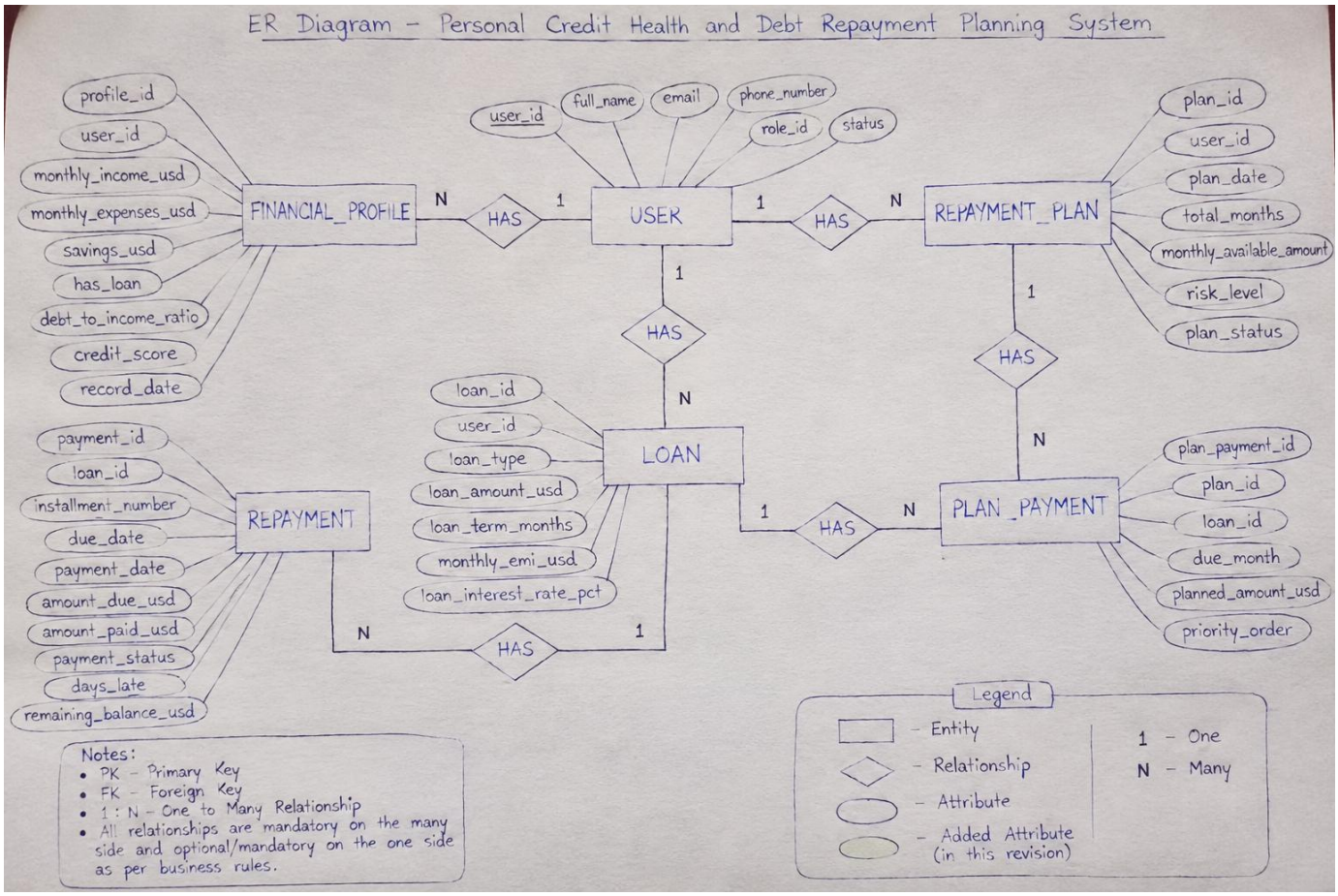
The absence of a centralized system makes it difficult to track outstanding loan amounts, monitor repayment deadlines, evaluate financial conditions and organize repayment priorities.

The proposed system addresses this problem by developing a relational database that stores personal financial information and maintains relationships between users, financial profiles, loans, repayments and repayment plans.

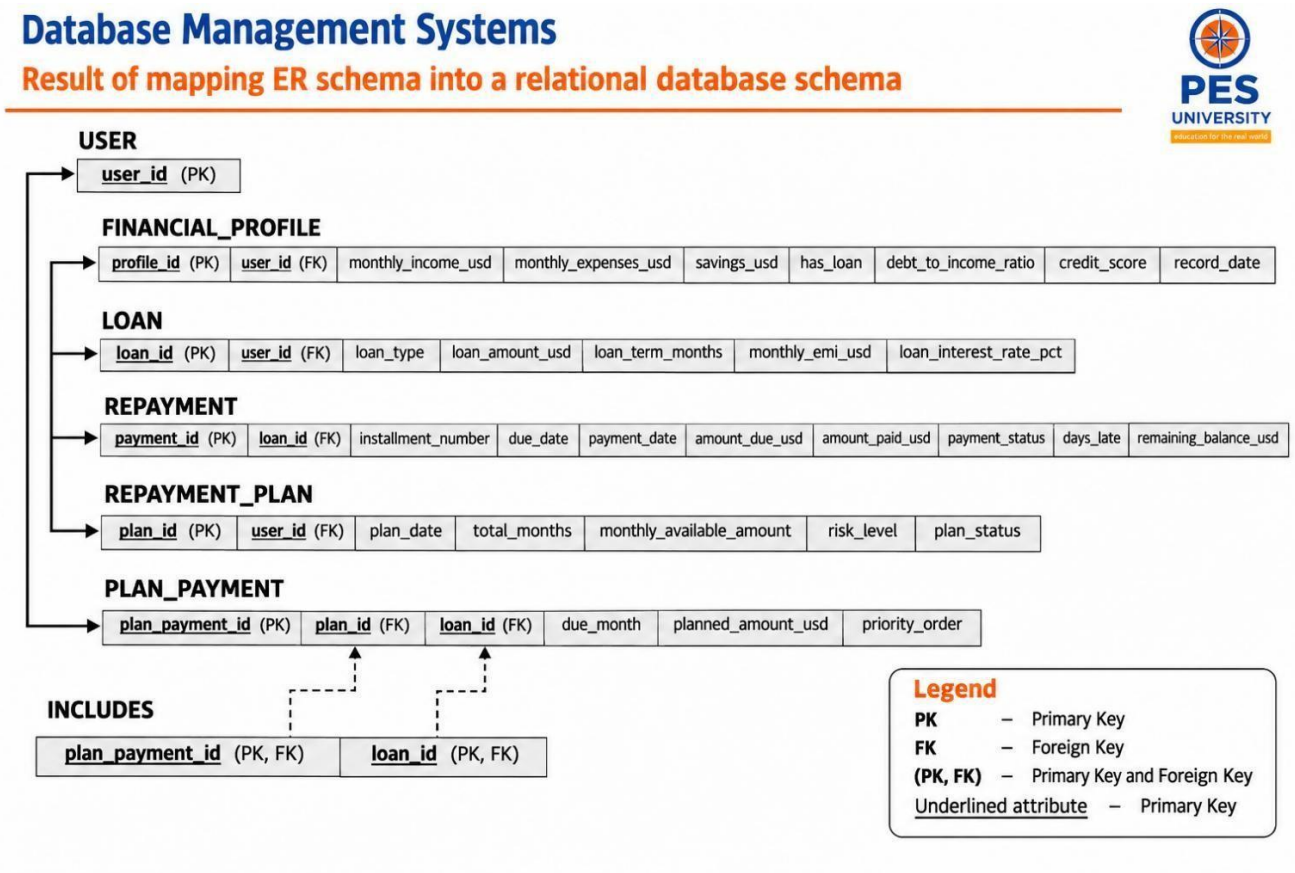
Proposed Solution:

The system provides a centralized MySQL database that allows financial records to be stored, updated, retrieved and analyzed using SQL operations. It also supports repayment planning and monitoring through related database tables.

3. ER MODEL


ER Diagram – Personal Credit Health and Debt Repayment Planning System
FINANCIAL_PROFILE
profile_id
user_id
monthly_income_usd
monthly_expenses_usd
savings_usd
has_loan
debt_to_income_ratio
credit_score
record_date
HAS
N
1
USER
user_id
full_name
email
phone_number
role_id
status
HAS
1
N
REPAYMENT_PLAN
plan_id
user_id
plan_date
total_months
monthly_available_amount
risk_level
plan_status
1
HAS
N
LOAN
loan_id
user_id
loan_type
loan_amount_usd
loan_term_months
monthly_emi_usd
loan_interest_rate_pct
1
HAS
N
PLAN_PAYMENT
plan_payment_id
plan_id
loan_id
due_month
planned_amount_usd
priority_order
REPAYMENT
payment_id
loan_id
installment_number
due_date
payment_date
amount_due_usd
amount_paid_usd
payment_status
days_late
remaining_balance_usd
N
HAS
1
Notes:
PK – Primary Key
FK – Foreign Key
1 : N – One to Many Relationship
All relationships are mandatory on the many side and optional/mandatory on the one side as per business rules.
Legend
– Entity
– Relationship
– Attribute
– Added Attribute (in this revision)
1 – One
N – Many

4. ER TO RELATIONAL MAPPING



5. DDL STATEMENTS

STATEMENTS WITH SCREEN SHOTS OF THE TABLE CREATION

5.1 **Database Creation and Selection**

```
mysql> CREATE DATABASE IF NOT EXISTS credit_health_db;
Query OK, 1 row affected, 1 warning (0.04 sec)

mysql> USE credit_health_db;
Database changed
mysql>
```

5.2 **Creation of USER Table**

```
mysql> CREATE TABLE IF NOT EXISTS `user` (
    ->     user_id VARCHAR(10) NOT NULL,
    ->     PRIMARY KEY (user_id)
    -> ) ENGINE=InnoDB;
Query OK, 0 rows affected, 1 warning (0.01 sec)

mysql> DESCRIBE `user`;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| user_id | varchar(10) | NO   | PRI | NULL    |       |
+---------+-------------+------+-----+---------+-------+
1 row in set (0.00 sec)
```

5.3 **Creation of FINANCIAL_PROFILE Table**

```
mysql> CREATE TABLE IF NOT EXISTS financial_profile (
    ->     profile_id INT NOT NULL AUTO_INCREMENT,
    ->     user_id VARCHAR(10) NOT NULL,
    ->     monthly_income_usd DECIMAL(12,2) NOT NULL,
    ->     monthly_expenses_usd DECIMAL(12,2) NOT NULL,
    ->     savings_usd DECIMAL(14,2) NOT NULL,
    ->     has_loan VARCHAR(3) NOT NULL,
    ->     debt_to_income_ratio DECIMAL(6,2) NOT NULL,
    ->     credit_score INT NOT NULL,
    ->     record_date DATE NOT NULL,
    ->     PRIMARY KEY (profile_id),
    ->     CONSTRAINT fk_financial_profile_user
    ->         FOREIGN KEY (user_id) REFERENCES `user` (user_id),
    ->     CONSTRAINT chk_profile_income CHECK (monthly_income_usd >= 0),
    ->     CONSTRAINT chk_profile_expenses CHECK (monthly_expenses_usd >= 0),
    ->     CONSTRAINT chk_profile_savings CHECK (savings_usd >= 0),
    ->     CONSTRAINT chk_profile_has_loan CHECK (has_loan IN ('Yes', 'No')),
    ->     CONSTRAINT chk_profile_credit_score CHECK (credit_score BETWEEN 300 AND 850)
    -> ) ENGINE=InnoDB;
Query OK, 0 rows affected (0.05 sec)

mysql>
mysql> DESCRIBE financial_profile;
+----------------------+---------------+------+-----+---------+----------------+
| Field                | Type          | Null | Key | Default | Extra          |
+----------------------+---------------+------+-----+---------+----------------+
| profile_id           | int           | NO   | PRI | NULL    | auto_increment |
| user_id              | varchar(10)   | NO   | MUL | NULL    |                |
| monthly_income_usd   | decimal(12,2) | NO   |     | NULL    |                |
| monthly_expenses_usd | decimal(12,2) | NO   |     | NULL    |                |
| savings_usd          | decimal(14,2) | NO   |     | NULL    |                |
| has_loan             | varchar(3)    | NO   |     | NULL    |                |
| debt_to_income_ratio | decimal(6,2)  | NO   |     | NULL    |                |
| credit_score         | int           | NO   |     | NULL    |                |
| record_date          | date          | NO   |     | NULL    |                |
+----------------------+---------------+------+-----+---------+----------------+
9 rows in set (0.00 sec)
```

5.4 **Creation of LOAN Table**

```
mysql> CREATE TABLE IF NOT EXISTS loan (
    ->     loan_id VARCHAR(10) NOT NULL,
    ->     user_id VARCHAR(10) NOT NULL,
    ->     loan_type VARCHAR(20) NOT NULL,
    ->     loan_amount_usd DECIMAL(14,2) NOT NULL,
    ->     loan_term_months INT NOT NULL,
    ->     monthly_emi_usd DECIMAL(12,2) NOT NULL,
    ->     loan_interest_rate_pct DECIMAL(6,2) NOT NULL,
    ->     PRIMARY KEY (loan_id),
    ->     CONSTRAINT fk_loan_user
    ->         FOREIGN KEY (user_id) REFERENCES `user` (user_id),
    ->     CONSTRAINT chk_loan_amount
    ->         CHECK (loan_amount_usd > 0),
    ->     CONSTRAINT chk_loan_term
    ->         CHECK (loan_term_months > 0),
    ->     CONSTRAINT chk_loan_emi
    ->         CHECK (monthly_emi_usd >= 0),
    ->     CONSTRAINT chk_loan_interest
    ->         CHECK (loan_interest_rate_pct >= 0)
    -> ) ENGINE=InnoDB;
Query OK, 0 rows affected, 1 warning (0.01 sec)

mysql>
mysql> DESCRIBE loan;
+-----------------------+---------------+------+-----+---------+-------+
| Field                 | Type          | Null | Key | Default | Extra |
+-----------------------+---------------+------+-----+---------+-------+
| loan_id               | varchar(10)   | NO   | PRI | NULL    |       |
| user_id               | varchar(10)   | NO   | MUL | NULL    |       |
| loan_type             | varchar(20)   | NO   |     | NULL    |       |
| loan_amount_usd       | decimal(14,2) | NO   |     | NULL    |       |
| loan_term_months      | int           | NO   |     | NULL    |       |
| monthly_emi_usd       | decimal(12,2) | NO   |     | NULL    |       |
| loan_interest_rate_pct | decimal(6,2)  | NO   |     | NULL    |       |
+-----------------------+---------------+------+-----+---------+-------+
7 rows in set (0.00 sec)
```

5.5 **Creation of REPAYMENT Table**

```
mysql> CREATE TABLE IF NOT EXISTS repayment (
    ->     payment_id VARCHAR(20) NOT NULL,
    ->     loan_id VARCHAR(10) NOT NULL,
    ->     installment_number INT NOT NULL,
    ->     due_date DATE NOT NULL,
    ->     payment_date DATE,
    ->     amount_due_usd DECIMAL(12,2) NOT NULL,
    ->     amount_paid_usd DECIMAL(12,2) NOT NULL DEFAULT 0.00,
    ->     payment_status VARCHAR(20) NOT NULL,
    ->     days_late INT NOT NULL DEFAULT 0,
    ->     remaining_balance_usd DECIMAL(14,2) NOT NULL,
    ->
    ->     PRIMARY KEY (payment_id),
    ->
    ->     CONSTRAINT fk_repayment_loan
    ->         FOREIGN KEY (loan_id) REFERENCES loan (loan_id),
    ->
    ->     CONSTRAINT chk_repayment_installment
    ->         CHECK (installment_number > 0),
    ->
    ->     CONSTRAINT chk_repayment_due
    ->         CHECK (amount_due_usd >= 0),
    ->
    ->     CONSTRAINT chk_repayment_paid
    ->         CHECK (amount_paid_usd >= 0),
    ->
    ->     CONSTRAINT chk_repayment_late
    ->         CHECK (days_late >= 0),
    ->
    ->     CONSTRAINT chk_repayment_balance
    ->         CHECK (remaining_balance_usd >= 0)
    -> ) ENGINE=InnoDB;
Query OK, 0 rows affected, 1 warning (0.01 sec)

mysql>
mysql> DESCRIBE repayment;
+----------------------+---------------+------+-----+---------+-------+
| Field                | Type          | Null | Key | Default | Extra |
+----------------------+---------------+------+-----+---------+-------+
| payment_id           | varchar(20)   | NO   | PRI | NULL    |       |
| loan_id              | varchar(10)   | NO   | MUL | NULL    |       |
| installment_number   | int           | NO   |     | NULL    |       |
| due_date             | date          | NO   | MUL | NULL    |       |
| payment_date         | date          | YES  |     | NULL    |       |
| amount_due_usd       | decimal(12,2) | NO   |     | NULL    |       |
| amount_paid_usd      | decimal(12,2) | NO   |     | 0.00    |       |
| payment_status       | varchar(20)   | NO   | MUL | NULL    |       |
| days_late            | int           | NO   |     | 0       |       |
| remaining_balance_usd | decimal(14,2) | NO   |     | NULL    |       |
+----------------------+---------------+------+-----+---------+-------+
10 rows in set (0.00 sec)
```

5.6 **Creation of REPAYMENT_PLAN Table**

```
mysql> CREATE TABLE IF NOT EXISTS repayment_plan (
    ->     plan_id INT NOT NULL AUTO_INCREMENT,
    ->     user_id VARCHAR(10) NOT NULL,
    ->     plan_date DATE NOT NULL,
    ->     total_months INT NOT NULL,
    ->     monthly_available_amount DECIMAL(12,2) NOT NULL,
    ->     risk_level VARCHAR(10) NOT NULL,
    ->     plan_status VARCHAR(15) NOT NULL DEFAULT 'Active',
    ->     PRIMARY KEY (plan_id),
    ->     CONSTRAINT fk_repayment_plan_user
    ->         FOREIGN KEY (user_id) REFERENCES `user` (user_id),
    ->     CONSTRAINT chk_plan_months
    ->         CHECK (total_months > 0),
    ->     CONSTRAINT chk_plan_available
    ->         CHECK (monthly_available_amount >= 0)
    -> ) ENGINE=InnoDB;
Query OK, 0 rows affected, 1 warning (0.01 sec)

mysql> DESCRIBE repayment_plan;
+--------------------------+---------------+------+-----+---------+----------------+
| Field                    | Type          | Null | Key | Default | Extra          |
+--------------------------+---------------+------+-----+---------+----------------+
| plan_id                  | int           | NO   | PRI | NULL    | auto_increment |
| user_id                  | varchar(10)   | NO   | MUL | NULL    |                |
| plan_date                | date          | NO   |     | NULL    |                |
| total_months             | int           | NO   |     | NULL    |                |
| monthly_available_amount | decimal(12,2) | NO   |     | NULL    |                |
| risk_level               | varchar(10)   | NO   |     | NULL    |                |
| plan_status              | varchar(15)   | NO   |     | Active  |                |
+--------------------------+---------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```

5.7 **Creation of PLAN_PAYMENT Table**

```
mysql> CREATE TABLE IF NOT EXISTS plan_payment (
    ->     plan_payment_id INT NOT NULL AUTO_INCREMENT,
    ->     plan_id INT NOT NULL,
    ->     loan_id VARCHAR(10) NOT NULL,
    ->     due_month DATE NOT NULL,
    ->     planned_amount_usd DECIMAL(12,2) NOT NULL,
    ->     priority_order INT NOT NULL,
    ->
    ->     PRIMARY KEY (plan_payment_id),
    ->
    ->     CONSTRAINT fk_plan_payment_plan
    ->         FOREIGN KEY (plan_id)
    ->         REFERENCES repayment_plan (plan_id),
    ->
    ->     CONSTRAINT fk_plan_payment_loan
    ->         FOREIGN KEY (loan_id)
    ->         REFERENCES loan (loan_id),
    ->
    ->     CONSTRAINT chk_planned_amount
    ->         CHECK (planned_amount_usd >= 0),
    ->
    ->     CONSTRAINT chk_priority_order
    ->         CHECK (priority_order > 0),
    ->
    ->     CONSTRAINT uq_plan_payment
    ->         UNIQUE (plan_id, loan_id, due_month)
    ->
    -> ) ENGINE=InnoDB;
Query OK, 0 rows affected, 1 warning (0.01 sec)

mysql>
mysql> DESCRIBE plan_payment;
+--------------------+---------------+------+-----+---------+----------------+
| Field              | Type          | Null | Key | Default | Extra          |
+--------------------+---------------+------+-----+---------+----------------+
| plan_payment_id    | int           | NO   | PRI | NULL    | auto_increment |
| plan_id            | int           | NO   | MUL | NULL    |                |
| loan_id            | varchar(10)   | NO   | MUL | NULL    |                |
| due_month          | date          | NO   |     | NULL    |                |
| planned_amount_usd | decimal(12,2) | NO   |     | NULL    |                |
| priority_order     | int           | NO   |     | NULL    |                |
+--------------------+---------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)
```

6. DML STATEMENTS

STATEMENTS WITH SCREEN SHOTS OF THE TABLE WITH INSERTED VALUES

6.1 **Insertion of USER Record**

```
mysql> INSERT INTO `user` (user_id)
    -> VALUES ('U03941')
    -> ON DUPLICATE KEY UPDATE user_id = VALUES(user_id);
Query OK, 0 rows affected, 1 warning (0.02 sec)

mysql> SELECT * FROM `user`
    -> WHERE user_id = 'U03941';
+---------+
| user_id |
+---------+
| U03941  |
+---------+
1 row in set (0.00 sec)
```

6.2 **Insertion of FINANCIAL_PROFILE Record**

```
mysql> INSERT INTO financial_profile
    ->     (user_id, monthly_income_usd, monthly_expenses_usd, savings_usd,
    ->      has_loan, debt_to_income_ratio, credit_score, record_date)
    -> SELECT 'U03941', 5000.00, 2500.00, 4000.00,
    ->        'Yes', 35.00, 720, '2026-08-24'
    -> WHERE NOT EXISTS (
    ->     SELECT 1 FROM financial_profile
    ->     WHERE user_id = 'U03941'
    ->     AND record_date = '2026-08-24'
    -> );
Query OK, 1 row affected (0.04 sec)
Records: 1  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM financial_profile
    -> WHERE user_id = 'U03941';
+------------+---------+-------------------+---------------------+-------------+----------+---------------------+--------------+-------------+
| profile_id | user_id | monthly_income_usd | monthly_expenses_usd | savings_usd | has_loan | debt_to_income_ratio | credit_score | record_date |
+------------+---------+-------------------+---------------------+-------------+----------+---------------------+--------------+-------------+
|       1573 | U03941  |           3936.35 |             1598.89 |   415121.73 | Yes      |                0.20 |          300 | 2022-11-28  |
|      16384 | U03941  |           5000.00 |             2500.00 |     4000.00 | Yes      |               35.00 |          720 | 2026-08-24  |
+------------+---------+-------------------+---------------------+-------------+----------+---------------------+--------------+-------------+
2 rows in set (0.00 sec)
```

6.3 **Insertion of LOAN Record**

```
mysql> INSERT INTO loan
    ->     (loan_id, user_id, loan_type, loan_amount_usd,
    ->      loan_term_months, monthly_emi_usd, loan_interest_rate_pct)
    -> VALUES
    ->     ('L03941', 'U03941', 'Personal', 10000.00, 24, 500.00, 8.50)
    ->
    -> ON DUPLICATE KEY UPDATE
    ->     user_id = 'U03941',
    ->     loan_type = 'Personal',
    ->     loan_amount_usd = 10000.00,
    ->     loan_term_months = 24,
    ->     monthly_emi_usd = 500.00,
    ->     loan_interest_rate_pct = 8.50;
Query OK, 2 rows affected (0.02 sec)

mysql>
mysql> SELECT *
    -> FROM loan
    -> WHERE loan_id = 'L03941';
+---------+---------+-----------+-----------------+------------------+-----------------+------------------------+
| loan_id | user_id | loan_type | loan_amount_usd | loan_term_months | monthly_emi_usd | loan_interest_rate_pct |
+---------+---------+-----------+-----------------+------------------+-----------------+------------------------+
| L03941  | U03941  | Personal  |        10000.00 |               24 |          500.00 |                   8.50 |
+---------+---------+-----------+-----------------+------------------+-----------------+------------------------+
1 row in set (0.00 sec)
```

6.4 **Insertion of REPAYMENT_PLAN Record**

```
mysql> INSERT INTO repayment_plan
    ->     (plan_id, user_id, plan_date, total_months,
    ->      monthly_available_amount, risk_level, plan_status)
    -> VALUES
    ->     (1, 'U03941', '2026-08-24', 12, 2339.46, 'High', 'Active')
    ->
    -> ON DUPLICATE KEY UPDATE
    ->     user_id = 'U03941',
    ->     plan_date = '2026-08-24',
    ->     total_months = 12,
    ->     monthly_available_amount = 2339.46,
    ->     risk_level = 'High',
    ->     plan_status = 'Active';
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> SELECT
    ->     plan_id,
    ->     user_id,
    ->     plan_date,
    ->     total_months,
    ->     monthly_available_amount,
    ->     risk_level,
    ->     plan_status
    -> FROM repayment_plan
    -> WHERE plan_id = 1;
+---------+---------+------------+--------------+--------------------------+------------+-------------+
| plan_id | user_id | plan_date  | total_months | monthly_available_amount | risk_level | plan_status |
+---------+---------+------------+--------------+--------------------------+------------+-------------+
|       1 | U03941  | 2026-08-24 |           12 |                  2339.46 | High       | Active      |
+---------+---------+------------+--------------+--------------------------+------------+-------------+
1 row in set (0.00 sec)
```

6.5 **Insertion of PLAN_PAYMENT Record**

```
mysql> INSERT INTO plan_payment
    ->     (plan_id, loan_id, due_month, planned_amount_usd, priority_order)
    -> VALUES
    ->     (1, 'L03941', '2026-09-01', 794.82, 1)
    ->
    -> ON DUPLICATE KEY UPDATE
    ->     planned_amount_usd = 794.82,
    ->     priority_order = 1;
Query OK, 1 row affected (0.06 sec)

mysql>
mysql> SELECT
    ->     plan_payment_id,
    ->     plan_id,
    ->     loan_id,
    ->     due_month,
    ->     planned_amount_usd,
    ->     priority_order
    -> FROM plan_payment
    -> WHERE plan_id = 1;
+-----------------+---------+---------+------------+--------------------+----------------+
| plan_payment_id | plan_id | loan_id | due_month  | planned_amount_usd | priority_order |
+-----------------+---------+---------+------------+--------------------+----------------+
|               5 |       1 | L03941  | 2026-09-01 |             794.82 |              1 |
+-----------------+---------+---------+------------+--------------------+----------------+
1 row in set (0.01 sec)
```

7. QUERIES

7.1 **Grouping Loans by Type Using Aggregate Functions**

```
mysql> SELECT
    ->     loan_type,
    ->     COUNT(*) AS total_loans,
    ->     SUM(loan_amount_usd) AS total_loan_amount,
    ->     AVG(loan_amount_usd) AS average_loan_amount
    -> FROM loan
    -> GROUP BY loan_type;
+-----------+-------------+-------------------+---------------------+
| loan_type | total_loans | total_loan_amount | average_loan_amount |
+-----------+-------------+-------------------+---------------------+
| Education |        3275 |      826544640.39 |       252380.042867 |
| Business  |        3261 |      808450183.75 |       247914.806424 |
| Car       |        3175 |      801443117.30 |       252423.029071 |
| Home      |        3284 |      809682258.38 |       246553.671857 |
+-----------+-------------+-------------------+---------------------+
4 rows in set (0.04 sec)
```

7.2 **Updating the Planned Payment Amount**

```
mysql> UPDATE plan_payment
    -> SET planned_amount_usd = 874.30
    -> WHERE plan_id = 1
    -> AND loan_id = 'L03941'
    -> AND due_month = '2026-09-01';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT *
    -> FROM plan_payment
    -> WHERE plan_id = 1;
+-----------------+---------+---------+------------+--------------------+----------------+
| plan_payment_id | plan_id | loan_id | due_month  | planned_amount_usd | priority_order |
+-----------------+---------+---------+------------+--------------------+----------------+
|               5 |       1 | L03941  | 2026-09-01 |             874.30 |              1 |
+-----------------+---------+---------+------------+--------------------+----------------+
1 row in set (0.00 sec)
```

7.3 **Deleting a Planned Payment Record**

```
mysql> USE credit_health_db;
Database changed
mysql>
mysql> DELETE FROM plan_payment
    -> WHERE plan_id = 1
    -> AND loan_id = 'L03941'
    -> AND due_month = '2026-09-01';
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT *
    -> FROM plan_payment
    -> WHERE plan_id = 1;
Empty set (0.00 sec)
```

7.4 **Retrieving Loan Details Using a Correlated Subquery**

```
mysql> SELECT
    ->     l.loan_id,
    ->     l.user_id,
    ->     l.loan_type,
    ->     l.loan_amount_usd,
    ->     (
    ->         SELECT COUNT(*)
    ->         FROM repayment r
    ->         WHERE r.loan_id = l.loan_id
    ->     ) AS total_repayments
    -> FROM loan l
    -> LIMIT 10;
+---------+---------+-----------+-----------------+------------------+
| loan_id | user_id | loan_type | loan_amount_usd | total_repayments |
+---------+---------+-----------+-----------------+------------------+
| L00002  | U00002  | Education |       146323.34 |               13 |
| L00004  | U00004  | Business  |        93242.37 |               13 |
| L00006  | U00006  | Car       |       407915.25 |               13 |
| L00007  | U00007  | Car       |       296057.60 |               13 |
| L00010  | U00010  | Education |       324197.37 |               13 |
| L00011  | U00011  | Education |       127703.79 |               13 |
| L00014  | U00014  | Car       |       404941.02 |               13 |
| L00021  | U00021  | Business  |       425717.20 |               13 |
| L00022  | U00022  | Home      |       180386.08 |               13 |
| L00023  | U00023  | Education |        78365.77 |               13 |
+---------+---------+-----------+-----------------+------------------+
10 rows in set (0.00 sec)
```

7.5 **Retrieving Financial Profiles Using a Nested Query**

```
mysql> SELECT
    ->     user_id,
    ->     credit_score,
    ->     monthly_income_usd,
    ->     monthly_expenses_usd,
    ->     savings_usd
    -> FROM financial_profile
    -> WHERE user_id IN
    -> (
    ->     SELECT user_id
    ->     FROM loan
    -> )
    -> LIMIT 10;
+---------+--------------+--------------------+----------------------+-------------+
| user_id | credit_score | monthly_income_usd | monthly_expenses_usd | savings_usd |
+---------+--------------+--------------------+----------------------+-------------+
| U00002  |          543 |            3531.73 |              2367.99 |   260869.10 |
| U00004  |          461 |            5894.88 |              4440.12 |   304815.51 |
| U00006  |          718 |            4051.00 |              2244.42 |   194901.41 |
| U00007  |          761 |             983.69 |               332.62 |    37522.56 |
| U00010  |          558 |            5216.18 |              4369.36 |   203320.50 |
| U00011  |          412 |            3677.43 |              2247.92 |   311546.91 |
| U00014  |          788 |            5546.28 |              3630.24 |   453101.38 |
| U00021  |          622 |            8237.77 |              7316.78 |    83622.72 |
| U00022  |          707 |            7093.07 |              3500.65 |   723648.99 |
| U00023  |          641 |            1694.87 |               604.03 |   118470.87 |
+---------+--------------+--------------------+----------------------+-------------+
10 rows in set (0.01 sec)
```

8. STORED PROCEDURES

8.1.1 Stored Procedure Creation – GetLoanRepaymentDetails

Purpose: Retrieves all repayment details associated with a particular loan ID.

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetLoanRepaymentDetails(
    ->     IN p_loan_id VARCHAR(10)
    -> )
    -> BEGIN
    ->     SELECT
    ->         loan_id,
    ->         installment_number,
    ->         due_date,
    ->         payment_date,
    ->         amount_due_usd,
    ->         amount_paid_usd,
    ->         payment_status,
    ->         days_late,
    ->         remaining_balance_usd
    ->     FROM repayment
    ->     WHERE loan_id = p_loan_id
    ->     ORDER BY installment_number;
    -> END //
Query OK, 0 rows affected (0.01 sec)
```

8.1.2 Stored Procedure Execution

```
mysql>
mysql> DELIMITER ;
mysql> CALL GetLoanRepaymentDetails('L03941');
+---------+--------------------+------------+--------------+----------------+-----------------+----------------+-----------+-----------------------+
| loan_id | installment_number | due_date   | payment_date | amount_due_usd | amount_paid_usd | payment_status | days_late | remaining_balance_usd |
+---------+--------------------+------------+--------------+----------------+-----------------+----------------+-----------+-----------------------+
| L03941  |                  1 | 2021-11-28 | 2021-11-28   |         794.82 |          794.82 | Paid           |         0 |             113662.03 |
| L03941  |                  2 | 2021-12-28 | 2021-12-28   |         794.82 |          794.82 | Paid           |         0 |             112867.21 |
| L03941  |                  3 | 2022-01-28 | 2022-01-28   |         794.82 |          794.82 | Paid           |         0 |             112072.39 |
| L03941  |                  4 | 2022-02-28 | 2022-03-01   |         794.82 |          794.82 | Paid Late      |         1 |             111277.57 |
| L03941  |                  5 | 2022-03-28 | 2022-03-28   |         794.82 |          794.82 | Paid           |         0 |             110482.75 |
| L03941  |                  6 | 2022-04-28 | 2022-04-28   |         794.82 |          794.82 | Paid           |         0 |             109687.93 |
| L03941  |                  7 | 2022-05-28 | 2022-05-28   |         794.82 |          794.82 | Paid           |         0 |             108893.11 |
| L03941  |                  8 | 2022-06-28 | 2022-06-28   |         794.82 |          794.82 | Paid           |         0 |             108098.29 |
| L03941  |                  9 | 2022-07-28 | 2022-08-24   |         794.82 |          794.82 | Paid Late      |        27 |             107303.47 |
| L03941  |                 10 | 2022-08-28 | 2022-09-19   |         794.82 |          794.82 | Paid Late      |        22 |             106508.65 |
| L03941  |                 11 | 2022-09-28 | NULL         |         794.82 |            0.00 | Missed         |        16 |             106508.65 |
| L03941  |                 12 | 2022-10-28 | 2022-10-28   |         794.82 |          794.82 | Paid           |         0 |             105713.83 |
| L03941  |                 13 | 2022-12-28 | NULL         |         794.82 |            0.00 | Upcoming       |         0 |             105713.83 |
+---------+--------------------+------------+--------------+----------------+-----------------+----------------+-----------+-----------------------+
13 rows in set (0.02 sec)
```

8.1.3 Stored Procedure Creation – GetOverdueRepayments

```
mysql>
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetOverdueRepayments(
    ->     IN p_as_of_date DATE
    -> )
    -> BEGIN
    ->     SELECT
    ->         r.payment_id,
    ->         r.loan_id,
    ->         l.user_id,
    ->         r.installment_number,
    ->         r.due_date,
    ->         r.amount_due_usd,
    ->         r.amount_paid_usd,
    ->         (r.amount_due_usd - r.amount_paid_usd)
    ->             AS unpaid_amount_usd,
    ->         GREATEST(
    ->             DATEDIFF(p_as_of_date, r.due_date), 0
    ->         ) AS overdue_days
    ->     FROM repayment AS r
    ->     INNER JOIN loan AS l
    ->         ON r.loan_id = l.loan_id
    ->     WHERE r.due_date < p_as_of_date
    ->       AND r.amount_paid_usd < r.amount_due_usd
    ->     ORDER BY r.due_date ASC
    ->     LIMIT 10;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql>
```

8.1.4 Execution of Stored Procedure – GetOverdueRepayments

```
mysql>
mysql> DELIMITER ;
mysql> CALL GetOverdueRepayments(CURDATE());
+------------+---------+---------+--------------------+------------+----------------+-----------------+--------------------+--------------+
| payment_id | loan_id | user_id | installment_number | due_date   | amount_due_usd | amount_paid_usd | unpaid_amount_usd | overdue_days |
+------------+---------+---------+--------------------+------------+----------------+-----------------+--------------------+--------------+
| P31073_01  | L31073  | U31073  |                  1 | 2020-07-24 |        7549.73 |            0.00 |            7549.73 |         2293 |
| P06091_01  | L06091  | U06091  |                  1 | 2020-07-26 |        5656.63 |            0.00 |            5656.63 |         2291 |
| P29764_01  | L29764  | U29764  |                  1 | 2020-07-26 |        2925.36 |            0.00 |            2925.36 |         2291 |
| P01966_01  | L01966  | U01966  |                  1 | 2020-07-27 |        3713.69 |            0.00 |            3713.69 |         2290 |
| P26647_01  | L26647  | U26647  |                  1 | 2020-07-29 |        6219.67 |            0.00 |            6219.67 |         2288 |
| P30283_01  | L30283  | U30283  |                  1 | 2020-07-30 |        4576.43 |            0.00 |            4576.43 |         2287 |
| P31534_01  | L31534  | U31534  |                  1 | 2020-07-30 |        8175.22 |            0.00 |            8175.22 |         2287 |
| P09498_01  | L09498  | U09498  |                  1 | 2020-07-31 |        3937.68 |            0.00 |            3937.68 |         2286 |
| P25658_01  | L25658  | U25658  |                  1 | 2020-07-31 |       22067.38 |            0.00 |           22067.38 |         2286 |
| P26321_01  | L26321  | U26321  |                  1 | 2020-08-01 |        6835.97 |            0.00 |            6835.97 |         2285 |
+------------+---------+---------+--------------------+------------+----------------+-----------------+--------------------+--------------+
10 rows in set (0.01 sec)

Query OK, 0 rows affected (0.10 sec)

mysql>
```

8.2 Trigger Creation and Verification

8.2.1 Creation of Calculate_Days_Late Trigger

```
mysql>
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER Calculate_Days_Late
    -> BEFORE INSERT ON repayment
    -> FOR EACH ROW
    -> BEGIN
    ->     SET NEW.days_late =
    ->         CASE
    ->             WHEN NEW.payment_date IS NOT NULL THEN
    ->                 GREATEST(
    ->                     DATEDIFF(NEW.payment_date, NEW.due_date),
    ->                     0
    ->                 )
    ->             ELSE
    ->                 GREATEST(
    ->                     DATEDIFF(CURDATE(), NEW.due_date),
    ->                     0
    ->                 )
    ->         END;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql>
```

8.2.1 Verification of Calculate_Days_Late Trigger

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO repayment
    -> (
    ->     payment_id,
    ->     loan_id,
    ->     installment_number,
    ->     due_date,
    ->     payment_date,
    ->     amount_due_usd,
    ->     amount_paid_usd,
    ->     payment_status,
    ->     days_late,
    ->     remaining_balance_usd
    -> )
    -> VALUES
    -> (
    ->     'TRG001',
    ->     'L03941',
    ->     99,
    ->     DATE_SUB(CURDATE(), INTERVAL 5 DAY),
    ->     NULL,
    ->     100.00,
    ->     0.00,
    ->     'Missed',
    ->     0,
    ->     100.00
    -> );
Query OK, 1 row affected (0.04 sec)

mysql>
mysql> SELECT
    ->     payment_id,
    ->     due_date,
    ->     payment_date,
    ->     days_late
    -> FROM repayment
    -> WHERE payment_id = 'TRG001';
+------------+------------+--------------+-----------+
| payment_id | due_date   | payment_date | days_late |
+------------+------------+--------------+-----------+
| TRG001     | 2026-10-29 | NULL         |         5 |
+------------+------------+--------------+-----------+
1 row in set (0.00 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.01 sec)

mysql>
```

8.2.2 Creation and Verification of Validate_Repayment_Amount Trigger

```
mysql>
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER Validate_Repayment_Amount
    -> BEFORE INSERT ON repayment
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.amount_paid_usd > NEW.amount_due_usd THEN
    ->
    ->         SIGNAL SQLSTATE '45000'
    ->             SET MESSAGE_TEXT =
    ->             'Invalid repayment: amount paid cannot exceed amount due';
    ->
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO repayment
    -> (
    ->     payment_id,
    ->     loan_id,
    ->     installment_number,
    ->     due_date,
    ->     payment_date,
    ->     amount_due_usd,
    ->     amount_paid_usd,
    ->     payment_status,
    ->     days_late,
    ->     remaining_balance_usd
    -> )
    -> VALUES
    -> (
    ->     'TRGAMT01',
    ->     'L03941',
    ->     98,
    ->     CURDATE(),
    ->     CURDATE(),
    ->     100.00,
    ->     150.00,
    ->     'Missed',
    ->     0,
    ->     0.00
    -> );
ERROR 1644 (45000): Invalid repayment: amount paid cannot exceed amount due
mysql>
```

8.2.3 Creation of Validate_Repayment_Date Trigger

```
mysql>
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER Validate_Repayment_Date
    -> BEFORE INSERT ON repayment
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.payment_date IS NOT NULL
    ->        AND NEW.payment_date > CURDATE() THEN
    ->
    ->         SIGNAL SQLSTATE '45000'
    ->             SET MESSAGE_TEXT =
    ->             'Invalid repayment: payment date cannot be in the future';
    ->
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO repayment
    -> (
    ->     payment_id,
    ->     loan_id,
    ->     installment_number,
    ->     due_date,
    ->     payment_date,
    ->     amount_due_usd,
    ->     amount_paid_usd,
    ->     payment_status,
    ->     days_late,
    ->     remaining_balance_usd
    -> )
    -> VALUES
    -> (
    ->     'TRGDATE01',
    ->     'L03941',
    ->     97,
    ->     CURDATE(),
    ->     DATE_ADD(CURDATE(), INTERVAL 1 DAY),
    ->     100.00,
    ->     50.00,
    ->     'Missed',
    ->     0,
    ->     50.00
    -> );
ERROR 1644 (45000): Invalid repayment: payment date cannot be in the future
mysql>
```

8.2.4 Creation of Validate_Repayment_Update Trigger

```
mysql>
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER Validate_Repayment_Update
    -> BEFORE UPDATE ON repayment
    -> FOR EACH ROW
    -> BEGIN
    ->
    ->     IF NEW.amount_paid_usd > NEW.amount_due_usd THEN
    ->
    ->         SIGNAL SQLSTATE '45000'
    ->             SET MESSAGE_TEXT =
    ->             'Invalid repayment update: amount paid cannot exceed amount due';
    ->
    ->     END IF;
    ->
    ->     IF NEW.payment_date IS NOT NULL
    ->        AND NEW.payment_date > CURDATE() THEN
    ->
    ->         SIGNAL SQLSTATE '45000'
    ->             SET MESSAGE_TEXT =
    ->             'Invalid repayment update: payment date cannot be in the future';
    ->
    ->     END IF;
    ->
    ->     SET NEW.days_late =
    ->         CASE
    ->             WHEN NEW.payment_date IS NOT NULL THEN
    ->                 GREATEST(
    ->                     DATEDIFF(NEW.payment_date, NEW.due_date), 0
    ->                 )
    ->             ELSE
    ->                 GREATEST(
    ->                     DATEDIFF(CURDATE(), NEW.due_date), 0
    ->                 )
    ->         END;
    ->
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql>
```

8.2.4.1 Verification of Days Late Calculation During Update

```
mysql>
mysql> INSERT INTO repayment
    -> (
    ->     payment_id,
    ->     loan_id,
    ->     installment_number,
    ->     due_date,
    ->     payment_date,
    ->     amount_due_usd,
    ->     amount_paid_usd,
    ->     payment_status,
    ->     days_late,
    ->     remaining_balance_usd
    -> )
    -> VALUES
    -> (
    ->     'TRGUPD02',
    ->     'L03941',
    ->     96,
    ->     DATE_SUB(CURDATE(), INTERVAL 5 DAY),
    ->     NULL,
    ->     100.00,
    ->     0.00,
    ->     'Missed',
    ->     0,
    ->     100.00
    -> );
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT
    ->     payment_id,
    ->     due_date,
    ->     payment_date,
    ->     payment_status,
    ->     days_late
    -> FROM repayment
    -> WHERE payment_id = 'TRGUPD02';
+------------+------------+--------------+----------------+-----------+
| payment_id | due_date   | payment_date | payment_status | days_late |
+------------+------------+--------------+----------------+-----------+
| TRGUPD02   | 2026-10-29 | NULL         | Missed         |         5 |
+------------+------------+--------------+----------------+-----------+
1 row in set (0.00 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.01 sec)

mysql>
```

8.2.4.2 Verification of Amount Validation During Update

```
mysql>
mysql> INSERT INTO repayment
    -> (
    ->     payment_id,
    ->     loan_id,
    ->     installment_number,
    ->     due_date,
    ->     payment_date,
    ->     amount_due_usd,
    ->     amount_paid_usd,
    ->     payment_status,
    ->     days_late,
    ->     remaining_balance_usd
    -> )
    -> VALUES
    -> (
    ->     'TRGB001',
    ->     'L03941',
    ->     97,
    ->     CURDATE(),
    ->     CURDATE(),
    ->     100.00,
    ->     150.00,
    ->     'Paid',
    ->     0,
    ->     0.00
    -> );
ERROR 1644 (45000): Invalid repayment: amount paid cannot exceed amount due
mysql>
mysql> SELECT *
    -> FROM repayment
    -> WHERE payment_id = 'TRGB001';
Empty set (0.00 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql>
```

8.2.4.3 Verification of Payment Date Validation During Update

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO repayment
    -> (
    ->     payment_id,
    ->     loan_id,
    ->     installment_number,
    ->     due_date,
    ->     payment_date,
    ->     amount_due_usd,
    ->     amount_paid_usd,
    ->     payment_status,
    ->     days_late,
    ->     remaining_balance_usd
    -> )
    -> VALUES
    -> (
    ->     'TRGC001',
    ->     'L03941',
    ->     98,
    ->     CURDATE(),
    ->     DATE_ADD(CURDATE(), INTERVAL 5 DAY),
    ->     100.00,
    ->     50.00,
    ->     'Pending',
    ->     0,
    ->     50.00
    -> );
ERROR 1644 (45000): Invalid repayment: payment date cannot be in the future
mysql>
mysql> SELECT *
    -> FROM repayment
    -> WHERE payment_id = 'TRGC001';
Empty set (0.00 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql>
```

8.2.4.4 Additional Amount Validation Test

```
mysql>
mysql> INSERT INTO repayment
    -> (
    ->     payment_id,
    ->     loan_id,
    ->     installment_number,
    ->     due_date,
    ->     payment_date,
    ->     amount_due_usd,
    ->     amount_paid_usd,
    ->     payment_status,
    ->     days_late,
    ->     remaining_balance_usd
    -> )
    -> VALUES
    -> (
    ->     'TRGD001',
    ->     'L03941',
    ->     95,
    ->     CURDATE(),
    ->     CURDATE(),
    ->     100.00,
    ->     50.00,
    ->     'Pending',
    ->     0,
    ->     50.00
    -> );
ERROR 3819 (HY000): Check constraint 'repayment_chk_6' is violated.
mysql>
mysql> UPDATE repayment
    -> SET amount_paid_usd = 150.00
    -> WHERE payment_id = 'TRGD001';
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql>
mysql> SELECT
    ->     payment_id,
    ->     amount_due_usd,
    ->     amount_paid_usd,
    ->     payment_status
    -> FROM repayment
    -> WHERE payment_id = 'TRGD001';
Empty set (0.00 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql>
```

8.2.4.5 Additional Payment Date Validation Test

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO repayment
    -> (
    ->     payment_id,
    ->     loan_id,
    ->     installment_number,
    ->     due_date,
    ->     payment_date,
    ->     amount_due_usd,
    ->     amount_paid_usd,
    ->     payment_status,
    ->     days_late,
    ->     remaining_balance_usd
    -> )
    -> VALUES
    -> (
    ->     'TRGE001',
    ->     'L03941',
    ->     94,
    ->     CURDATE(),
    ->     NULL,
    ->     100.00,
    ->     0.00,
    ->     'Pending',
    ->     0,
    ->     100.00
    -> );
ERROR 3819 (HY000): Check constraint 'repayment_chk_6' is violated.
mysql>
mysql> UPDATE repayment
    -> SET payment_date = DATE_ADD(CURDATE(), INTERVAL 5 DAY)
    -> WHERE payment_id = 'TRGE001';
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql>
mysql> SELECT
    ->     payment_id,
    ->     due_date,
    ->     payment_date,
    ->     payment_status
    -> FROM repayment
    -> WHERE payment_id = 'TRGE001';
Empty set (0.00 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql>
```

8.2.4.6 Verification of Successful Repayment Update

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO repayment
    -> (
    ->     payment_id,
    ->     loan_id,
    ->     installment_number,
    ->     due_date,
    ->     payment_date,
    ->     amount_due_usd,
    ->     amount_paid_usd,
    ->     payment_status,
    ->     days_late,
    ->     remaining_balance_usd
    -> )
    -> VALUES
    -> (
    ->     'TRGF001',
    ->     'L03941',
    ->     93,
    ->     DATE_SUB(CURDATE(), INTERVAL 5 DAY),
    ->     NULL,
    ->     100.00,
    ->     0.00,
    ->     'Missed',
    ->     0,
    ->     100.00
    -> );
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> -- FIRST SELECT: Before UPDATE
mysql> SELECT
    ->     payment_id,
    ->     due_date,
    ->     payment_date,
    ->     payment_status,
    ->     days_late
    -> FROM repayment
    -> WHERE payment_id = 'TRGF001';
+------------+------------+--------------+----------------+------------+
| payment_id | due_date   | payment_date | payment_status | days_late |
+------------+------------+--------------+----------------+------------+
| TRGF001    | 2026-10-29 | NULL         | Missed         |          5 |
+------------+------------+--------------+----------------+------------+
1 row in set (0.00 sec)

mysql>
mysql> -- UPDATE
mysql> UPDATE repayment
    -> SET payment_date = DATE_SUB(CURDATE(), INTERVAL 2 DAY),
    ->     amount_paid_usd = 100.00,
    ->     payment_status = 'Paid',
    ->     remaining_balance_usd = 0.00
    -> WHERE payment_id = 'TRGF001';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> -- SECOND SELECT: After UPDATE
mysql> SELECT
    ->     payment_id,
    ->     due_date,
    ->     payment_date,
    ->     payment_status,
    ->     days_late
    -> FROM repayment
    -> WHERE payment_id = 'TRGF001';
+------------+------------+--------------+----------------+------------+
| payment_id | due_date   | payment_date | payment_status | days_late |
+------------+------------+--------------+----------------+------------+
| TRGF001    | 2026-10-29 | 2026-11-01   | Paid           |          3 |
+------------+------------+--------------+----------------+------------+
1 row in set (0.00 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.01 sec)

mysql>
```

8.3 Creation and Verification of Views

8.3.1 Creation and Verification of Loan Summary View

```
mysql> USE credit_health_db;
Database changed
mysql>
mysql> CREATE OR REPLACE VIEW vw_loan_summary AS
    -> SELECT
    ->     l.loan_id,
    ->     l.user_id,
    ->     l.loan_type,
    ->     l.loan_amount_usd,
    ->     l.monthly_emi_usd,
    ->     COUNT(r.payment_id) AS repayment_count,
    ->     COALESCE(SUM(r.remaining_balance_usd), 0)
    ->         AS remaining_balance_usd
    -> FROM loan l
    -> LEFT JOIN repayment r
    ->     ON l.loan_id = r.loan_id
    -> GROUP BY
    ->     l.loan_id,
    ->     l.user_id,
    ->     l.loan_type,
    ->     l.loan_amount_usd,
    ->     l.monthly_emi_usd;
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> -- Display only 10 loan records
mysql> SELECT *
    -> FROM vw_loan_summary
    -> ORDER BY loan_id
    -> LIMIT 10;
+---------+---------+-----------+-----------------+-----------------+-----------------+-----------------------+
| loan_id | user_id | loan_type | loan_amount_usd | monthly_emi_usd | repayment_count | remaining_balance_usd |
+---------+---------+-----------+-----------------+-----------------+-----------------+-----------------------+
| L00002  | U00002  | Education |       146323.34 |         4953.50 |              13 |            1505923.42 |
| L00004  | U00004  | Business  |        93242.37 |         4926.57 |              13 |             798318.93 |
| L00006  | U00006  | Car       |       407915.25 |        38219.81 |              13 |            2346758.39 |
| L00007  | U00007  | Car       |       296057.60 |         6773.29 |              13 |            3239152.70 |
| L00010  | U00010  | Education |       324197.37 |         6782.37 |              13 |            3604152.51 |
| L00011  | U00011  | Education |       127703.79 |         1818.01 |              13 |            1496528.37 |
| L00014  | U00014  | Car       |       404941.02 |        10823.31 |              13 |            4290135.36 |
| L00021  | U00021  | Business  |       425717.20 |        13258.10 |              13 |            4632772.80 |
| L00022  | U00022  | Home      |       180386.08 |         2286.50 |              13 |            2168958.54 |
| L00023  | U00023  | Education |        78365.77 |         1315.94 |              13 |             906900.11 |
+---------+---------+-----------+-----------------+-----------------+-----------------+-----------------------+
10 rows in set (1.97 sec)
```

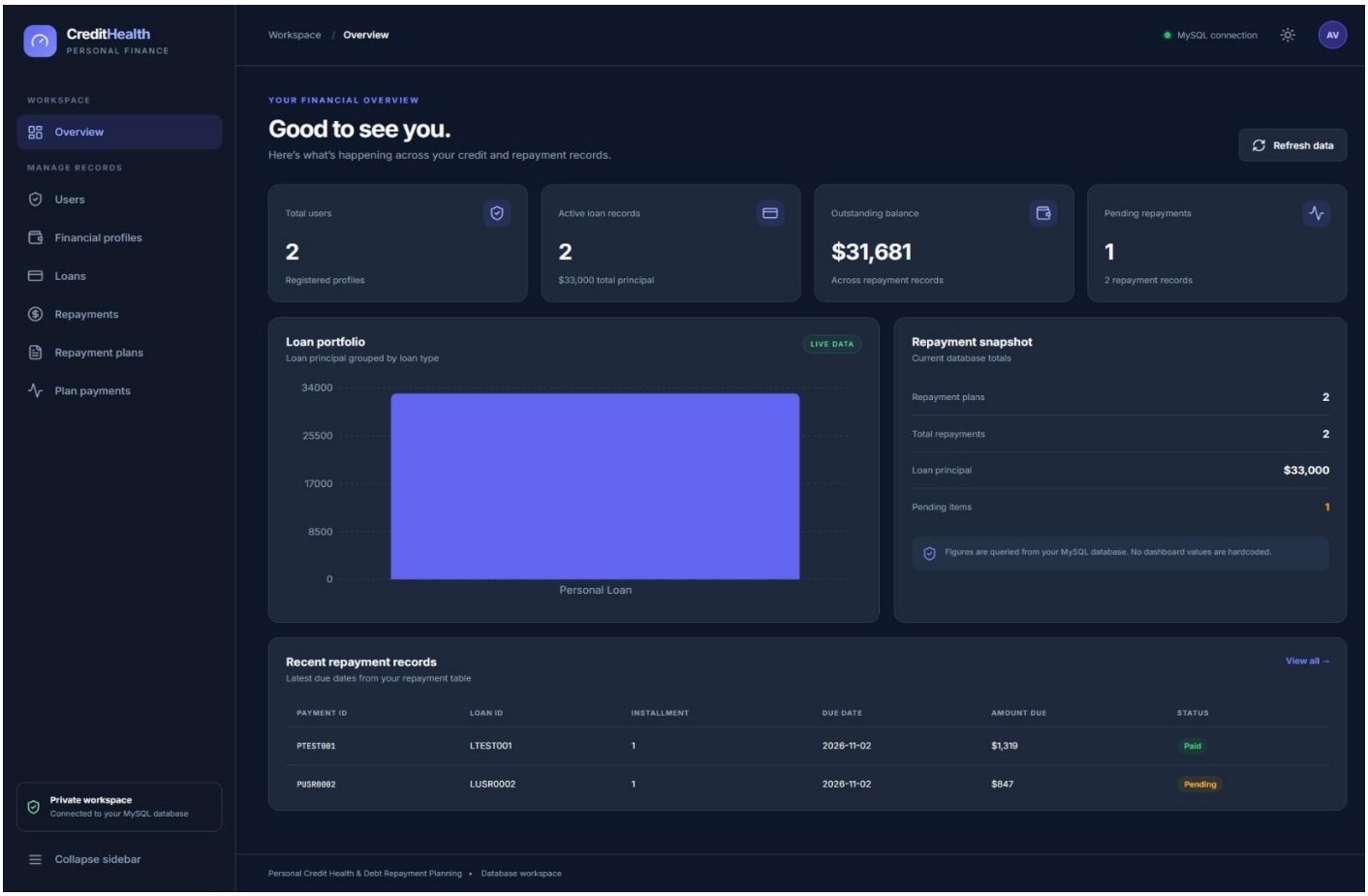
8.3.2 Creation and Verification of Repayment Status View

```
mysql> USE credit_health_db;
Database changed
mysql>
mysql> -- 8.3.2 Create the repayment status view
mysql> CREATE OR REPLACE VIEW vw_repayment_status AS
    -> SELECT
    ->     payment_id,
    ->     loan_id,
    ->     installment_number,
    ->     due_date,
    ->     payment_date,
    ->     amount_due_usd,
    ->     amount_paid_usd,
    ->     payment_status,
    ->     days_late,
    ->     remaining_balance_usd
    -> FROM repayment;
Query OK, 0 rows affected (0.02 sec)

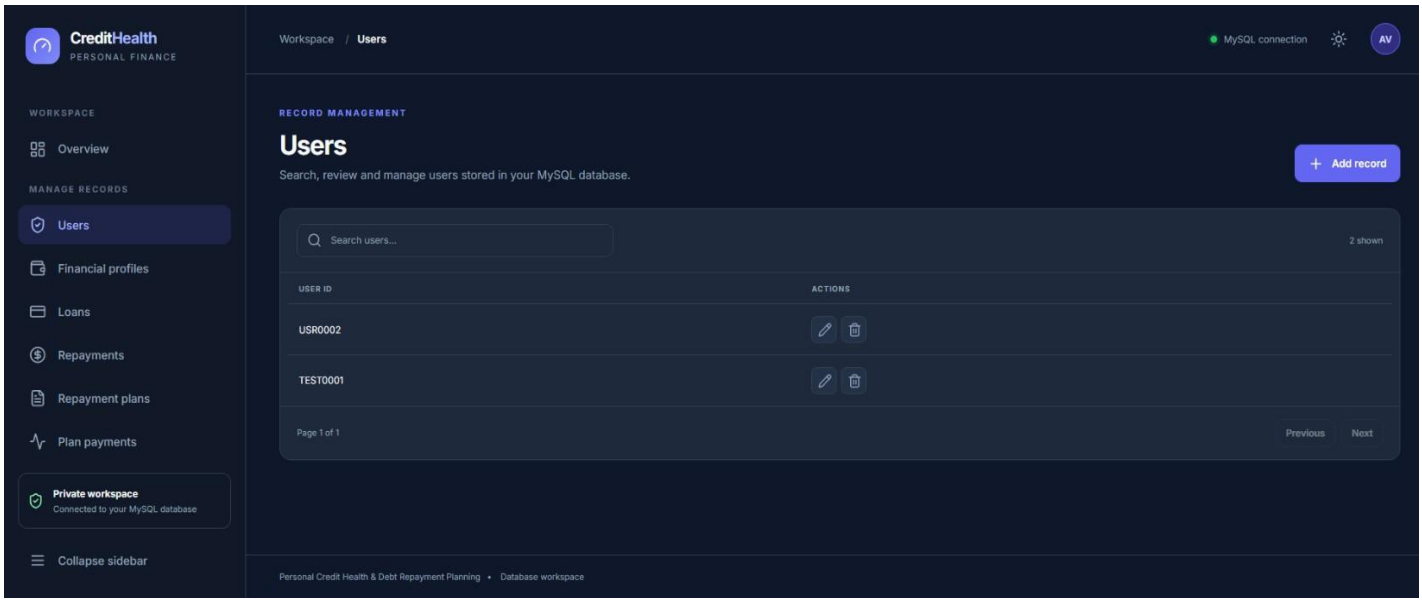
mysql>
mysql> -- Verify the view
mysql> SELECT *
    -> FROM vw_repayment_status
    -> ORDER BY loan_id, installment_number
    -> LIMIT 10;
+------------+---------+-------------------+------------+--------------+----------------+----------------+----------------+-----------+-----------------------+
| payment_id | loan_id | installment_number | due_date   | payment_date | amount_due_usd | amount_paid_usd | payment_status | days_late | remaining_balance_usd |
+------------+---------+-------------------+------------+--------------+----------------+----------------+----------------+-----------+-----------------------+
| P00002_01  | L00002  |                 1 | 2021-02-13 | 2021-03-09   |        4953.50 |         4953.50 | Paid Late      |        24 |             141369.84 |
| P00002_02  | L00002  |                 2 | 2021-03-13 | 2021-03-13   |        4953.50 |         4953.50 | Paid           |         0 |             136416.34 |
| P00002_03  | L00002  |                 3 | 2021-04-13 | 2021-05-03   |        4953.50 |         4953.50 | Paid Late      |        20 |             131462.84 |
| P00002_04  | L00002  |                 4 | 2021-05-13 | 2021-05-13   |        4953.50 |         4953.50 | Paid           |         0 |             126509.34 |
| P00002_05  | L00002  |                 5 | 2021-06-13 | 2021-06-13   |        4953.50 |         4953.50 | Paid           |         0 |             121555.84 |
| P00002_06  | L00002  |                 6 | 2021-07-13 | NULL         |        4953.50 |            0.00 | Missed         |        25 |             121555.84 |
| P00002_07  | L00002  |                 7 | 2021-08-13 | 2021-09-07   |        4953.50 |         4953.50 | Paid Late      |        25 |             116602.34 |
| P00002_08  | L00002  |                 8 | 2021-09-13 | 2021-09-30   |        4953.50 |         4953.50 | Paid Late      |        17 |             111648.84 |
| P00002_09  | L00002  |                 9 | 2021-10-13 | 2021-10-13   |        4953.50 |         4953.50 | Paid           |         0 |             106695.34 |
| P00002_10  | L00002  |                10 | 2021-11-13 | 2021-11-13   |        4953.50 |         4953.50 | Paid           |         0 |             101741.84 |
+------------+---------+-------------------+------------+--------------+----------------+----------------+----------------+-----------+-----------------------+
10 rows in set (0.34 sec)
```

9.FRONT END DEVELOPEMNT

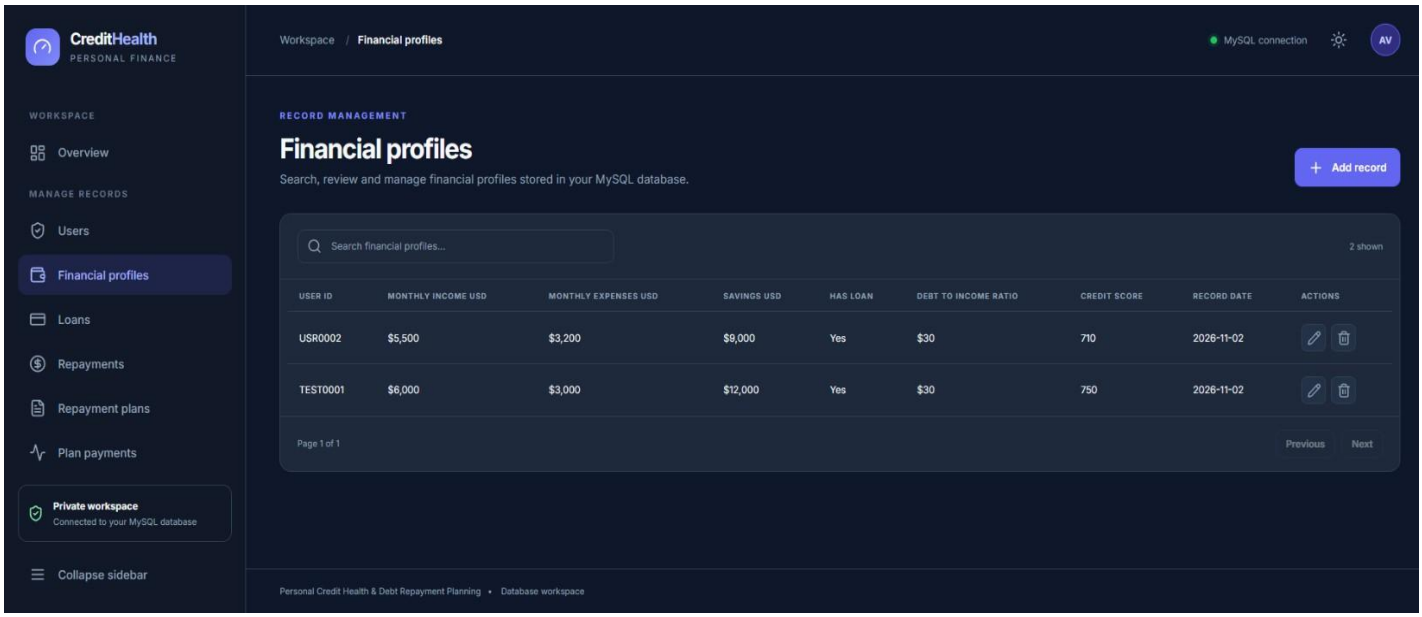
9.1 Financial Overview Dashboard



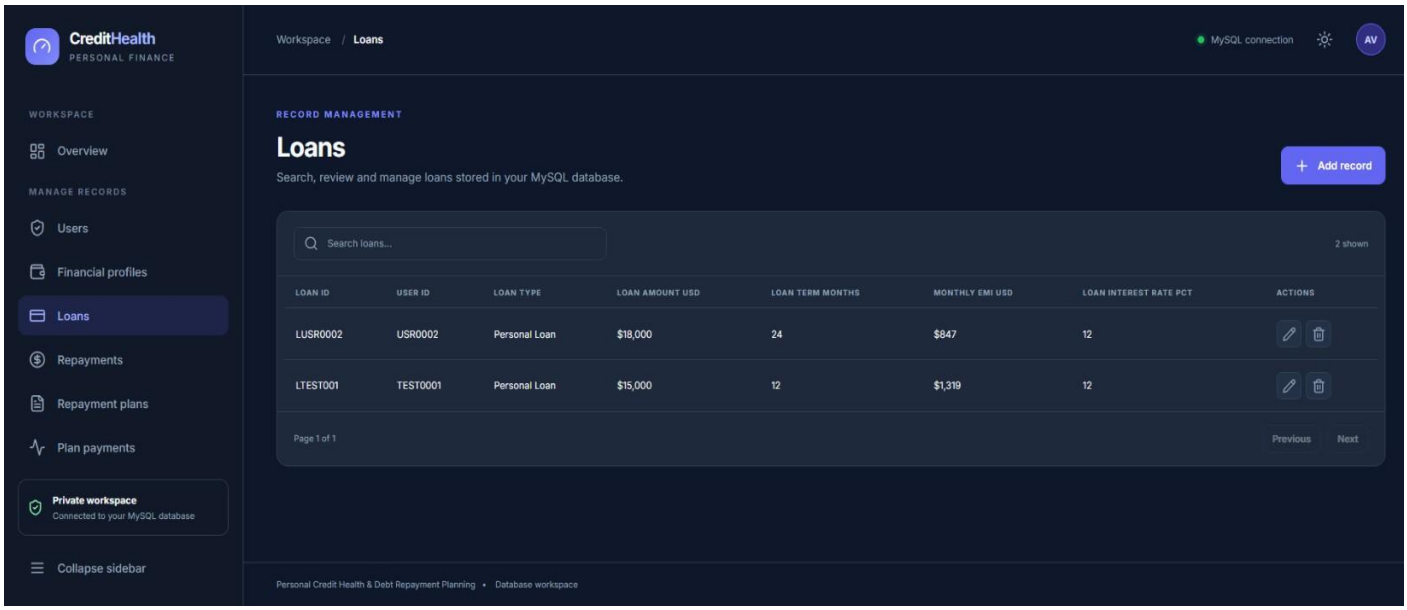
9.2 User Management



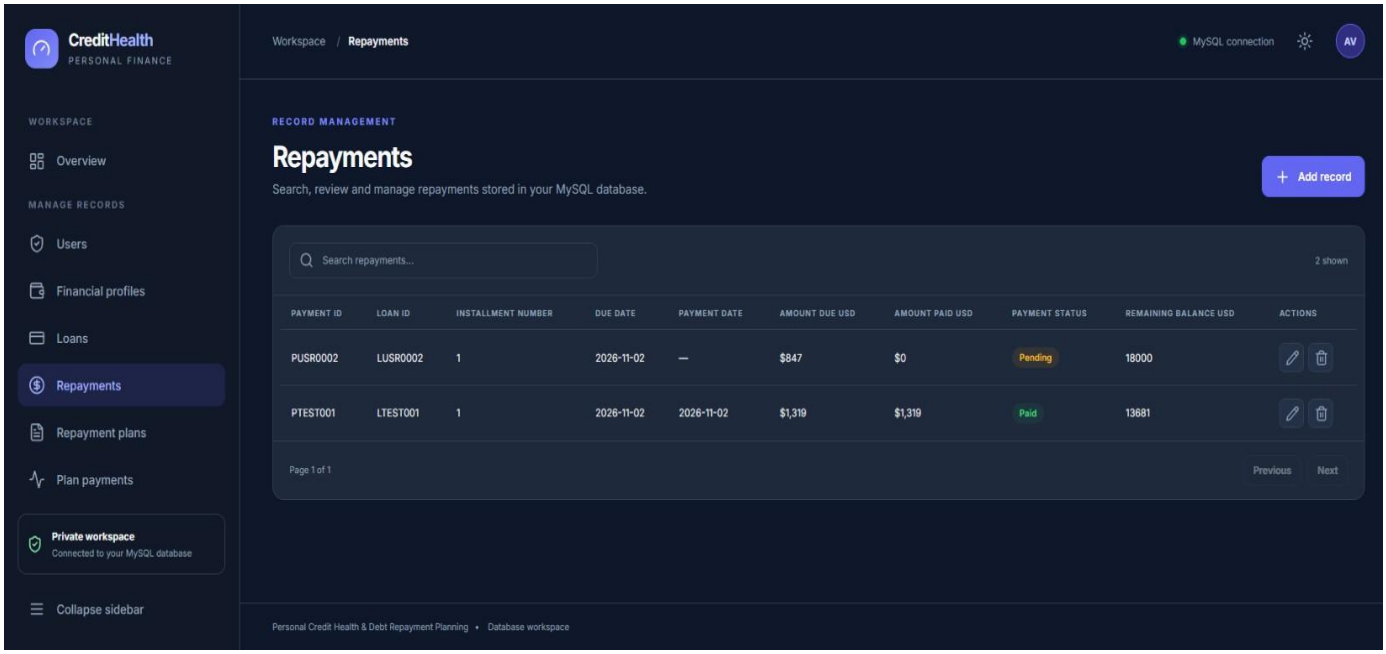
9.3 Financial Profile Management



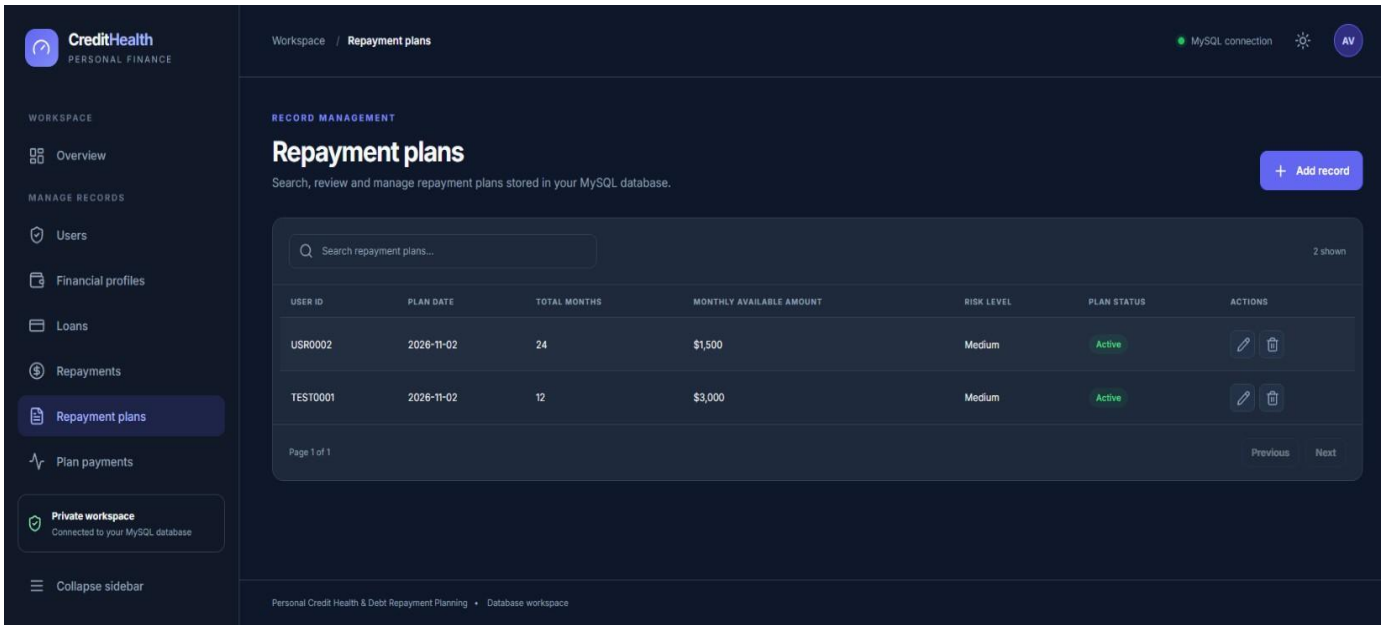
9.4 Loan Management



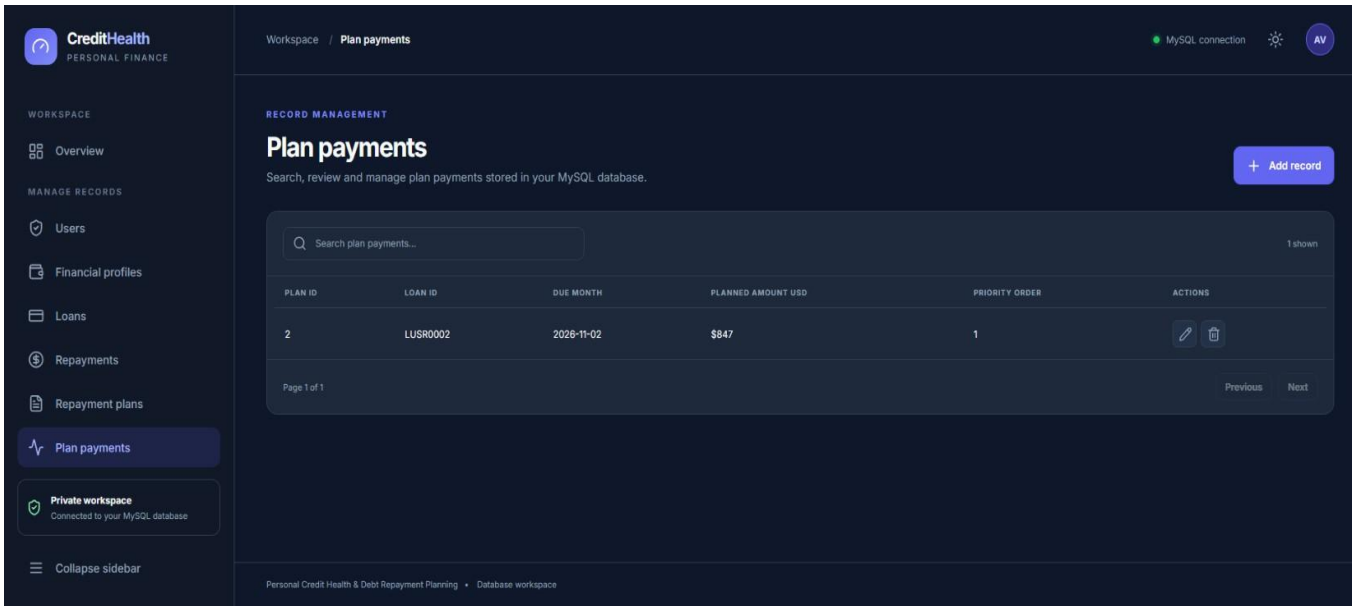
9.5 Repayment Management



9.6 Repayment Plan Management



9.7 Plan Payment Management



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3. Flask Documentation – https://flask.palletsprojects.com/
4. React Documentation – https://react.dev/
5. Vite Documentation – https://vite.dev/guide/
6. PyMySQL Documentation – https://pymysql.readthedocs.io/
7. MDN Web Docs – https://developer.mozilla.org/
8. GitHub Documentation – https://docs.github.com/

GitHub Link:

https://github.com/amoghaar2006/Personal-Credit-Health-Debt-Repayment-System-V2